

BHAGYASREE ROY

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PROFESSIONAL SUMMARY

MCA student (CHRIST University, expected 2027) and BCA graduate (CGPA 9.52/10) with hands-on experience across data analysis, machine learning, full-stack and mobile development, and Generative AI integration. Comfortable owning a feature or a research question end to end: requirements, design, implementation, testing, documentation, and, in one case, peer review. Background spans a software development internship, a client-facing consulting internship, seven independent technical projects, one accepted conference paper, and a growing set of self-directed certifications spanning cloud infrastructure, data science, and web development.

EDUCATION

CHRIST (Deemed to be University)

Master of Computer Applications (MCA), Ongoing

Bengaluru, India

Jun 2025 – May 2027 (Expected)

CHRIST (Deemed to be University)

Bachelor of Computer Applications (BCA), CGPA: 9.52/10 (83%)

Bengaluru, India

Jul 2022 – Apr 2025

St. Joseph's Convent

High School Diploma, PCM-C

Chandannagar, India

2020 – 2022

PROFESSIONAL EXPERIENCE

Junior Software Developer Intern

IOLITE Technologies (P) Ltd.

Nov 2024 – Mar 2025

Bengaluru, India

- Contributed to end-to-end feature development of an online degree platform: requirements analysis, technical specification, and delivery across frontend and backend layers.
- Supported data integration and validation workflows across platform modules, identifying inconsistencies and ensuring data quality across the pipeline.
- Conducted end-to-end testing and QA cycles, identified functional defects and data discrepancies, and worked on performance optimisation.
- Collaborated cross-functionally with design, content, and QA teams to verify feature correctness and integrate course materials.
- Documented system behaviour, defect findings, and resolution outcomes to support knowledge transfer and process improvement.

Deputy Creative Head

Christ Consulting

Sep 2023 – Jul 2024

Bengaluru, India

- Engaged with clients to understand vision, goals, and requirements; structured project briefs and delivery frameworks for creative and consulting engagements.
- Managed timelines, milestones, and deliverables across multiple concurrent client engagements, consistently meeting deadlines.
- Reviewed design work for quality and consistency against client expectations.
- Preceded by a Student Intern role at the same organisation, Jul – Sep 2023.

PROJECTS

MAGE, Multi-Agent AI System | *Python, LangGraph, Anthropic API, Multimodal LLMs*

Jun 2025 – Present

- Building a multi-agent pipeline with LangGraph orchestration; own multimodal data ingestion/schema normalisation and goal-conditioned chart selection and generation modules.
- Integrating generative AI APIs for autonomous agent decision-making across heterogeneous data inputs.

AI-Generated Artwork Forensic Detection | *Python, Scikit-learn, OpenCV, SVM, Random Forest, MLP*

Jan 2025 – Present

- Engineered 12 handcrafted forensic features (noise, texture, frequency-domain) from a 9,000-image dataset spanning 10 artistic styles, for a 3-class attribution task (human art vs. standard diffusion vs. latent diffusion).
- Trained and evaluated SVM, Random Forest, and MLP classifiers: 90.3% accuracy (0.9576 AUC) on binary detection, 89.6% accuracy (0.9762 AUC) on 3-class attribution.
- Co-authored the resulting paper, "Forensic Feature Analysis for AI-Generated Artwork Detection: Distinguishing Human Art from Standard and Latent Diffusion Models," with Chandra J. and Arokia Paul Rajan R. (Dept. of Computer Science, CHRIST University). Accepted at the 7th Congress on Intelligent Systems (CIS 2026), NIT Warangal and Soft Computing Research Society, Sep 5–6, 2026; to be published in Springer's *Lecture Notes in Networks and Systems* (Scopus-indexed). See Publications below.

Procto, AI-Powered Proctoring System | *TypeScript, React, Vite, Express, PostgreSQL, Docker*

Jan 2025 – Mar 2025

- Architected a full-stack platform: React/Vite frontend, Express REST API backend, PostgreSQL data layer, and Docker Compose containerisation.
- Designed a shared TypeScript type schema across frontend, backend, and API layers to enforce data consistency and eliminate integration bugs.
- Integrated AI-driven behavioural monitoring; defined data flows for real-time analysis pipelines and manually verified API request/response flows pre-deployment.
- Filed a copyright application for the platform (registration pending).

Silver Analytics Dashboard | *Python, Plotly/Dash, Pandas, GeoPandas, Shapefile*

Aug 2024 – Oct 2024

- Preprocessed and validated time-series silver price data; performed geographic joins across state-level India datasets via GeoPandas.

- Built an interactive Plotly/Dash dashboard with choropleth maps and trend charts for multi-dimensional trend and regional demand analysis.

Consumer Shopping Behaviour Analysis | *Python, Pandas, Scikit-learn, Matplotlib, Seaborn* Jun 2024 – Aug 2024

- Collected, cleaned, and audited a structured consumer dataset; ran exploratory data analysis to uncover purchasing patterns, segment customers, and surface anomalies.
- Applied ML classification for customer segmentation, evaluated model performance, and produced stakeholder-ready visualisations for retail decision-making.

Christite, Student Mobile App | *React Native, Expo, Appwrite* Aug 2024 – Oct 2024

- Delivered a multi-feature university app: secure authentication, timetable management, and real-time push notifications via a cloud backend.

HEAL, Mental Health Web Platform | *HTML, CSS, JavaScript, PHP, JSON* Feb 2024 – Apr 2024

- Designed and implemented full-stack features, including personalised health tracking modules and secure communication channels between users and professionals.

PUBLICATIONS

- Roy, B., Chandra, J., and Arokia Paul Rajan, R. “Forensic Feature Analysis for AI-Generated Artwork Detection: Distinguishing Human Art from Standard and Latent Diffusion Models.” Accepted at the 7th Congress on Intelligent Systems (CIS 2026), organised by NIT Warangal and the Soft Computing Research Society, Sep 5–6, 2026. To be published in Springer’s *Lecture Notes in Networks and Systems* (Scopus-indexed).

TECHNICAL SKILLS

Languages: Python, SQL, Java, C, TypeScript, JavaScript

AI & Automation: Generative AI APIs, LLMs, LangGraph, Anthropic API, Google Gemini API

Data Science & ML: Pandas, NumPy, Scikit-learn, Statsmodels, Matplotlib, Seaborn, GeoPandas, OpenCV, WEKA, Power BI, Excel

Databases: PostgreSQL, MySQL, SQL Query Optimisation, Appwrite, Prisma ORM, Database Management

Dev & Frameworks: React, React Native, Expo, Express.js, Vite, PHP, HTML/CSS, Android Studio

DevOps & Tools: Docker, Docker Compose, Git/GitHub, Jupyter, Google Colab, AWS (S3, EC2), Google Cloud (Compute Engine, Load Balancing)

Concepts: EDA, Feature Engineering, Model Evaluation & Optimisation, Data Cleaning/Auditing/Validation, Data Visualisation, REST APIs, Webhooks, ETL/ELT, System Design, Requirements Analysis, Agile, AI Workflow Automation

SPOKEN LANGUAGES

Bengali (Native/Bilingual) • English (Native/Bilingual) • Hindi (Limited Working) • German (Elementary)

CERTIFICATIONS & TRAINING

- **Google Cloud Engineering Certificate**, Google Cloud Career Launchpad (2026)
- **Google Cybersecurity Professional Certificate**, Coursera/Google, 8 courses (Jun 2023)
- **AWS Cloud Foundations**, AWS Academy (2024)
- **The Complete 2024 Web Development Bootcamp**, Udemy, 61.5 hours, instructor Dr. Angela Yu (Apr 2024)
- **Machine Learning (using Python)**, Simplilearn SkillUp (May 2024)
- **Introduction to Cloud Computing**, Simplilearn SkillUp (Sep 2024)
- **Data Engineering 101: The Beginner’s Guide**, Udemy (Jun 2026)
- **Digital Marketing**, CHRIST University (2023)
- **Public Speaking**, NPTEL/SWAYAM, IIT Roorkee, Elite, 78% (Jul–Oct 2023)
- **Intermediate Python**, DataCamp (Jun 2025)
- **Intermediate Object-Oriented Programming in Python**, DataCamp (Jun 2025)
- **Data Manipulation with pandas**, DataCamp (Jul 2025)
- **Winning a Kaggle Competition in Python**, DataCamp (Jul 2025)
- **Python for Beginners: Data Structures**, Coursera Project Network (Jun 2025)
- **Introduction to JavaScript: The Basics**, Coursera Project Network (Jul 2025)
- **Use Canva to Create Desktop and Mobile-Friendly Web Pages**, Coursera Project Network

Google Cloud Skill Boost badges (supporting the Google Cloud Engineering Certificate; actively pursuing the Associate Cloud Engineer path, 4,066 Google Skills points as of Aug 2026):

- Google Cloud Fundamentals: Core Infrastructure
- Essential Google Cloud Infrastructure: Foundation
- Essential Google Cloud Infrastructure: Core Services
- Elastic Google Cloud Infrastructure: Scaling and Automation
- Implementing Cloud Load Balancing for Compute Engine
- Build an ACE Certification Study Guide

LEADERSHIP & EXTRACURRICULAR

Club Manager

Nov 2021 – Jan 2022

UABLE PTE LTD (Remote)

India

- Collected and analysed participant feedback data to identify trends and drive iterative improvements in event quality and engagement.
- Coordinated multi-stakeholder events, managing logistics, budgets, and communications for 10+ virtual events.

Student Council Member, Sustainable Development Goals (SDG) Cell

Sep 2023 – Jul 2024

CHRIST University

Bengaluru, India

Head of Creatives

2022 – 2025

SAMAGRA (CS Association), CHRIST University

Bengaluru, India

Core Member

WIST Women Coding Club, CHRIST University

Bengaluru, India

Member

SWO Art Cultural Team, CHRIST University

Bengaluru, India

Student Council Representative

2025 – 2026

CHRIST University

Bengaluru, India

- Elected liaison between the student body and administration.

HONORS & AWARDS

- **First Runner-Up, OpenWeaver & Christ Makeathon**, 48-hour Digital Product Development challenge (Sep 7, 2023)
- **Academic Excellence**, CGPA 9.52/10, BCA, CHRIST University
- **Certificate of Participation, Women Who Master Hackathon**, all-women national initiative by Logitech & Aspire For Her, via Unstop (Jul 27, 2026)